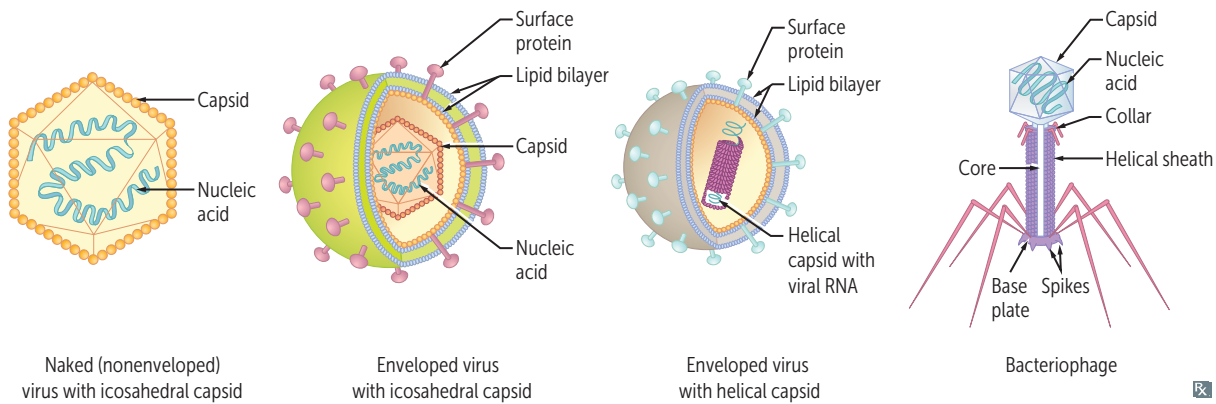


► MICROBIOLOGY—VIROLOGY

Viral structure—general features



Viral genetics

Recombination

Exchange of genes between 2 chromosomes by crossing over within regions of significant base sequence homology.



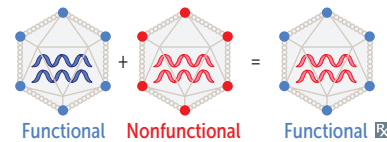
Reassortment

When viruses with segmented genomes (eg, influenza virus) exchange genetic material. For example, the 2009 novel H1N1 influenza A pandemic emerged via complex viral reassortment of genes from human, swine, and avian viruses. Has potential to cause antigenic shift. Reassortment of genome segments.



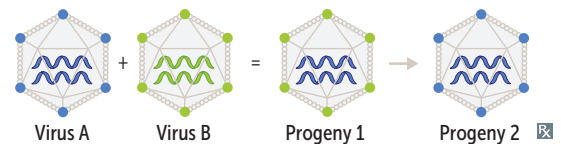
Complementation

When 1 of 2 viruses that infect the cell has a mutation that results in a nonfunctional protein, the nonmutated virus “complements” the mutated one by making a functional protein that serves both viruses. For example, hepatitis D virus requires the presence of replicating hepatitis B virus to supply HBsAg, the envelope protein for HDV.



Phenotypic mixing

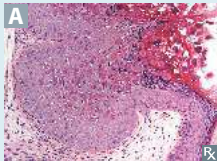
Occurs with simultaneous infection of a cell with 2 viruses. For progeny 1, genome of virus A can be partially or completely coated (forming pseudovirion) with the surface proteins of virus B. Type B protein coat determines the tropism (infectivity) of the hybrid virus. Progeny from subsequent infection of a cell by progeny 1 will have a type A coat that is encoded by its type A genetic material.



Viral genomes		
Naked nucleic acids of most dsDNA viruses (except poxviruses and HBV) and $\oplus$ strand ssRNA viruses are infectious. Naked nucleic acids of $\ominus$ strand ssRNA and dsRNA viruses are not infectious because they lack the required polymerases to replicate. Virions of $\ominus$ strand ssRNA viruses carry RNA-dependent RNA polymerases to transcribe $\ominus$ strand to $\oplus$.		
	CHARACTERISTICS	MNEMONIC
DNA viruses	All have dsDNA genomes (like our cells) except Parvoviridae (ssDNA).	Part of a virus
	All are linear except papilloma-, polyoma-, and hepadnaviruses (circular).	
RNA viruses	All have ssRNA genomes except Reoviridae (dsRNA).	Repeato-virus
	$\oplus$ stranded ($\approx$ mRNA): retro- , toga- , flavi- , corona- , hepe- , calici- , and picornaviruses .	While at a retro toga party, I drank flavored Corona and ate hippie California pickles .
	$\ominus$ stranded: arena- , bunya- , paramyxo- , orthomyxo- , filo- , and rhabdoviruses .	Always bring polymerase or fail replication .
	Segmented: Bunya- , Orthomyxo- , Arena- , and Reoviruses .	BOAR
Viral envelopes		
	Generally, enveloped viruses acquire their envelopes from plasma membrane when they exit from cell. Exceptions include herpesviruses, which acquire envelopes from nuclear membrane.	Enveloped DNA viruses (herpesvirus , hepadnavirus , poxvirus) have helpful protection .

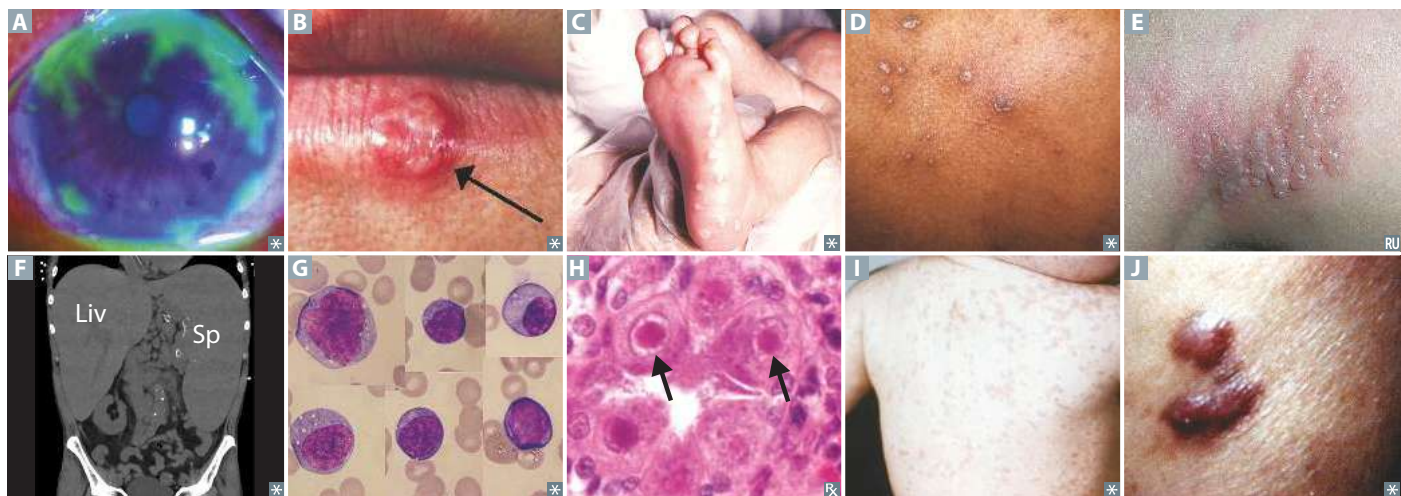
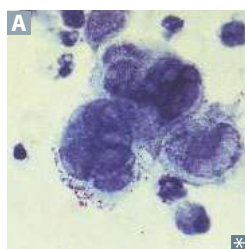
DNA viruses

All are icosahedral and replicate in the nucleus (except poxvirus). “**Pox** is out of the **box** (nucleus).”

VIRAL FAMILY	ENVELOPE	DNA STRUCTURE	MEDICAL IMPORTANCE
Herpesviruses	Yes	DS and linear	See Herpesviruses entry
Poxvirus 	Yes	DS and linear (largest DNA virus)	Smallpox eradicated world wide by use of the live-attenuated vaccine Cowpox (“milkmaid blisters”) Molluscum contagiosum —flesh-colored papule with central umbilication; keratinocytes contain molluscum bodies A
Hepadnavirus	Yes	Partially DS and circular	HBV: ▪ Acute or chronic hepatitis ▪ Not a retrovirus but has reverse transcriptase
Adenovirus 	No	DS and linear	Febrile pharyngitis B —sore throat Acute hemorrhagic cystitis Pneumonia Conjunctivitis—“pink eye” Gastroenteritis Myocarditis
Papillomavirus	No	DS and circular	HPV—warts, cancer (cervical, anal, penile, or oropharyngeal); serotypes 1, 2, 6, 11 associated with warts; serotypes 16, 18 associated with cancer
Polyomavirus	No	DS and circular	JC virus—progressive multifocal leukoencephalopathy (PML) in immunocompromised patients (eg, HIV) BK virus—transplant patients, commonly targets kidney JC : Junky C erebrum; BK : Bad K idney
Parvovirus	No	SS and linear (smallest DNA virus; <i>parvus</i> = small)	B19 virus—aplastic crises in sickle cell disease, “slapped cheek” rash in children (erythema infectiosum, or fifth disease); infects RBC precursors and endothelial cells → RBC destruction → hydrops fetalis and death in fetus, pure RBC aplasia and rheumatoid arthritis–like symptoms in adults

Herpesviruses Enveloped, DS, and linear viruses. Recent data suggest both HSV-1 and HSV-2 can affect both genital and extragenital areas.

VIRUS	ROUTE OF TRANSMISSION	CLINICAL SIGNIFICANCE	NOTES
Herpes simplex virus-1	Respiratory secretions, saliva	Gingivostomatitis, keratoconjunctivitis A , herpes labialis (cold sores) B , herpetic whitlow on finger, temporal lobe encephalitis, esophagitis, erythema multiforme. Responsible for a growing percentage of herpes genitalis.	Most commonly latent in trigeminal ganglia Most common cause of sporadic encephalitis, can present as altered mental status, seizures, and/or aphasia
Herpes simplex virus-2	Sexual contact, perinatal	Herpes genitalis, neonatal herpes C	Most commonly latent in sacral ganglia Viral meningitis more common with HSV-2 than with HSV-1
Varicella-zoster virus (HHV-3)	Respiratory secretions, contact with fluid from vesicles	Varicella-zoster (chickenpox D , shingles E), encephalitis, pneumonia Most common complication of shingles is post-herpetic neuralgia	Latent in dorsal root or trigeminal ganglia; CN V ₁ branch involvement can cause herpes zoster ophthalmicus
Epstein-Barr virus (HHV-4)	Respiratory secretions, saliva; also called “kissing disease,” (common in teens, young adults)	Mononucleosis —fever, hepatosplenomegaly F , pharyngitis, and lymphadenopathy (especially posterior cervical nodes); avoid contact sports until resolution due to risk of splenic rupture Associated with lymphomas (eg, endemic Burkitt lymphoma), nasopharyngeal carcinoma (especially Asian adults), lymphoproliferative disease in transplant patients	Infects B cells through CD 21 , “Must be 21 to drink B eer in a B arr” Atypical lymphocytes on peripheral blood smear G —not infected B cells but reactive cytotoxic T cells ⊕ Monospot test—heterophile antibodies detected by agglutination of sheep or horse RBCs Use of amoxicillin (eg, for presumed strep pharyngitis) can cause maculopapular rash
Cytomegalovirus (HHV-5)	Congenital, transfusion, sexual contact, saliva, urine, transplant	Mononucleosis (⊖ Monospot) in immunocompetent patients; infection in immunocompromised, especially pneumonia in transplant patients; esophagitis; colitis; AIDS retinitis (“ sight omegalovirus”): hemorrhage, cotton-wool exudates, vision loss Congenital CMV	Infected cells have characteristic “owl eye” intranuclear inclusions H Latent in mononuclear cells
Human herpes-viruses 6 and 7	Saliva	Roseola infantum (exanthem subitum): high fevers for several days that can cause seizures, followed by diffuse macular rash (starts on trunk then spreads to extremities) I ; usually seen in children < 2 years old	Roseola : fever first, Rosy (rash) later Self-limited illness HHV-7—less common cause of roseola
Human herpesvirus 8	Sexual contact	Kaposi sarcoma (neoplasm of endothelial cells). Seen in HIV/AIDS and transplant patients. Dark/violaceous plaques or nodules J representing vascular proliferations	Can also affect GI tract and lungs

Herpesviruses (continued)**HSV identification**

PCR of skin lesions is test of choice.

CSF PCR for herpes encephalitis.

Tzanck test (outdated)—a smear of an opened skin vesicle to detect multinucleated giant cells **A** commonly seen in HSV-1, HSV-2, and VZV infection.

Intranuclear eosinophilic Cowdry A inclusions also seen with HSV-1, HSV-2, VZV.

Receptors used by viruses

VIRUS	RECEPTOR(S)
CMV	Integrins (heparan sulfate)
EBV	CD21
HIV	CD4, CXCR4, CCR5
Parvovirus B19	P antigen on RBCs
Rabies	Nicotinic AChR
Rhinovirus	ICAM-1 (I CAMe to see the rhino)
SARS-CoV-2	ACE2

RNA viruses			
All replicate in the cy toplasm (except retrovirus and infl uenza virus). “ Retro flu is outta cyt (sight).”			
VIRAL FAMILY	ENVELOPE	RNA STRUCTURE	MEDICAL IMPORTANCE
Reoviruses	No	DS linear Multisegmented	Rotavirus—important cause of diarrhea in young children; may be fatal.
Picornaviruses	No	SS ⊕ linear	P oliavirus—polio-Salk/Sabin vaccines—IPV/OPV E chovirus—aseptic meningitis R hinovirus—“common cold” C oxsackievirus—aseptic meningitis; herpangina (mouth blisters, fever); hand, foot, and mouth disease; myocarditis; pericarditis HAV —acute viral hepatitis PERCH
Hepevirus	No	SS ⊕ linear	HEV
Caliciviruses	No	SS ⊕ linear	Norovirus—viral gastroenteritis
Flaviviruses	Yes	SS ⊕ linear	HCV Yellow fever ^a Dengue ^a West Nile virus ^a —meningoencephalitis, acute asymmetric flaccid paralysis Zika virus ^a
Togaviruses	Yes	SS ⊕ linear	Toga CREW — C hikungunya virus ^a (co-infection with dengue virus can occur), R ubella (formerly a togavirus), E astern and W estern equine encephalitis ^a
Matonavirus	Yes	SS ⊕ linear	Rubella
Retroviruses	Yes	SS ⊕ linear	Have reverse transcriptase HTLV—T-cell leukemia HIV—AIDS
Coronaviruses	Yes	SS ⊕ linear	“Common cold,” SARS, COVID-19, MERS
Orthomyxoviruses	Yes	SS ⊖ linear Multisegmented	Influenza virus
Paramyxoviruses	Yes	SS ⊖ linear	PaRaMyxovirus : P arainfluenza—croup RSV —bronchiolitis in babies M easles, M umps
Pneumoviruses	Yes	SS ⊖ linear	RSV—bronchiolitis in babies
Rhabdoviruses	Yes	SS ⊖ linear	Rabies
Filoviruses	Yes	SS ⊖ linear	Ebola/Marburg virus disease—often fatal.
Arenaviruses	Yes	SS ⊕ and ⊖ circular Multisegmented	LCMV—lymphocytic choriomeningitis virus Lassa fever encephalitis—spread by rodents
Bunyaviruses	Yes	SS ⊖ circular Multisegmented	California encephalitis ^a Sandfly/Rift Valley fevers ^a Crimean-Congo hemorrhagic fever ^a Hantavirus—hemorrhagic fever, pneumonia
Delta virus	Yes	SS ⊖ circular	HDV is “ D efective”; requires presence of HBV to replicate

SS, single-stranded; DS, double-stranded; ⊕, positive sense; ⊖, negative sense; ^a= **arbo**virus, **ar**thropod **bo**rne (mosquitoes, ticks).

Picornavirus

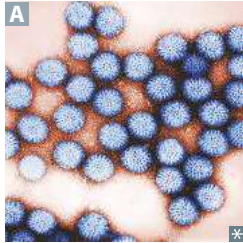
Includes **P**oliovirus, **E**chovirus, **R**hinovirus, **C**oxsackievirus, and **H**AV. RNA is translated into 1 large polypeptide that is cleaved by virus-encoded proteases into functional viral proteins. Poliovirus, echovirus, and coxsackievirus are enteroviruses and can cause aseptic (viral) meningitis.

Pico**RNA**virus = small **RNA** virus.
PERCH on a “**peak**” (**pico**).

Rhinovirus

A picornavirus. Nonenveloped RNA virus. Cause of common cold; > 100 serologic types. Acid labile—destroyed by stomach acid; therefore, does not infect the GI tract (unlike the other picornaviruses).

Rhino has a runny **nose**.

Rotavirus

Segmented dsRNA virus (a reovirus) **A**. Most important global cause of infantile gastroenteritis. Major cause of acute diarrhea in the United States during winter, especially in day care centers, kindergartens. Villous destruction with atrophy leads to ↓ absorption of Na^+ and loss of K^+ .

Rotavirus = **right out the anus**.
CDC recommends routine vaccination of all infants except those with a history of intussusception (rare adverse effect of rotavirus vaccination) or SCID.

Influenza viruses

Orthomyxoviruses. Enveloped, $\ominus$ ssRNA viruses with segmented genome. Contain hemagglutinin (binds sialic acid and promotes viral entry) and neuraminidase (promotes progeny virion release) antigens. Patients at risk for fatal bacterial superinfection, most commonly *S aureus*, *S pneumoniae*, and *H influenzae*. Treatment: supportive +/- neuraminidase inhibitor (eg, oseltamivir, zanamivir).

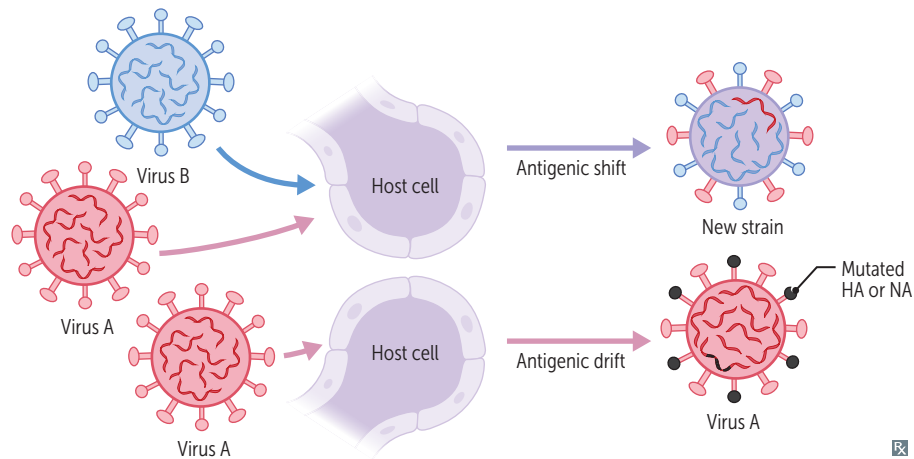
Hemagglutinin: lets the virus **in**
 Neuraminidase: sends the virus **away**
 Reformulated vaccine (“the flu shot”) contains viral strains most likely to appear during the flu season, due to the virus’ rapid genetic change. Killed viral vaccine is most frequently used. Live attenuated vaccine contains temperature-sensitive mutant that replicates in the nose but not in the lung; administered intranasally. Sudden **shift** is more deadly than **gradual drift**.

Genetic/antigenic shift

Infection of 1 cell by 2 different segmented viruses (eg, swine influenza and human influenza viruses) → RNA segment reassortment → dramatically different virus (genetic shift) → major global outbreaks (pandemics).

Genetic/antigenic drift

Random mutation in hemagglutinin (HA) or neuraminidase (NA) genes → minor changes in HA or NA protein (drift) occur frequently → local seasonal outbreaks (epidemics).

**Rubella virus**

A matonavirus. Causes rubella, formerly called German (3-day) measles. Fever, postauricular and other lymphadenopathy, arthralgias, and fine, maculopapular rash that starts on face and spreads centrifugally to involve trunk and extremities **A**.

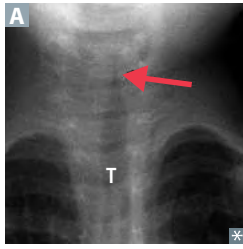
Causes mild disease in children but serious congenital disease (a TORCH infection). Congenital rubella findings include classic triad of sensorineural deafness, cataracts, and patent ductus arteriosus. “Blueberry muffin” appearance may be seen due to dermal extramedullary hematopoiesis.

Paramyxoviruses

Paramyxoviruses cause disease in children. They include those that cause parainfluenza (croup), mumps, measles, RSV, and human metapneumovirus. All subtypes can cause respiratory tract infection (bronchiolitis, pneumonia) in infants. All contain surface F (fusion) protein, which causes respiratory epithelial cells to fuse and form multinucleated cells. Palivizumab (monoclonal antibody against F protein) prevents pneumonia caused by RSV infection in premature infants.

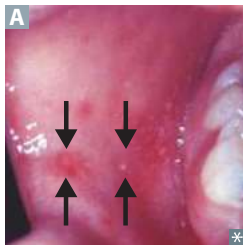
Palivizumab for **p**aramyxovirus (RSV) **p**rophylaxis in **p**reemies.

Acute laryngotracheobronchitis



Also called croup. Caused by parainfluenza viruses. Virus membrane contains hemagglutinin (binds sialic acid and promotes viral entry) and neuraminidase (promotes progeny virion release) antigens. Results in a “seal-like” barking cough and inspiratory stridor. Narrowing of upper trachea and subglottis leads to characteristic steeple sign on x-ray **A**.

Measles (rubeola) virus



Usual presentation involves prodromal fever with cough, coryza, and conjunctivitis, then eventually Koplik spots (bright red spots with blue-white center on buccal mucosa **A**), followed 1–2 days later by a maculopapular rash that starts at the head/neck and spreads downward.

Lymphadenitis with Warthin-Finkeldey giant cells (fused lymphocytes) in a background of paracortical hyperplasia. Possible sequelae:

- Subacute sclerosing panencephalitis (SSPE): personality changes, dementia, autonomic dysfunction, death (occurs years later)
- Encephalitis (1:1000): symptoms appear within few days of rash
- Giant cell pneumonia (rare except in immunosuppressed)

4 **C**'s of measles:

Cough

Coryza

Conjunctivitis

“**C**”oplik spots

Vitamin A supplementation can reduce morbidity and mortality from measles, particularly in malnourished children.

Pneumonia is the most common cause of measles-associated death in children.

Mumps virus



Uncommon due to effectiveness of MMR vaccine.

Symptoms: **P**arotitis **A**, **O**rchitis (inflammation of testes), aseptic **M**eningitis, and **P**ancreatitis. Can cause sterility (especially after puberty).

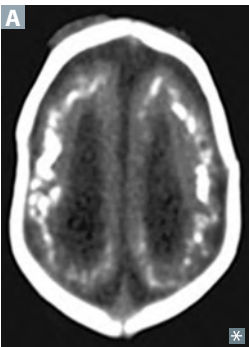
Mumps makes your parotid glands and testes as big as **POM-Poms**.

Arboviruses transmitted by *Aedes* mosquitoes

	Chikungunya virus	Dengue virus
VIRUS TYPE	Alphavirus/togavirus.	Flavivirus.
SYMPTOMS	High fever, maculopapular rash, headache, lymphadenopathy, and inflammatory polyarthrititis. Arthralgias are more commonly reported (vs dengue); joint swelling is highly specific for Chikungunya. Thrombocytopenia, leukopenia, and hemorrhagic manifestations are less common.	Dengue fever: fever, rash, headache, myalgias, arthralgias, retro-orbital pain, neutropenia. Dengue hemorrhagic fever: dengue fever + bleeding and plasma leakage due to severe thrombocytopenia and RBC perturbations. Most common if infected with a different serotype after initial infection due to antibody-dependent enhancement of disease. May progress to dengue shock syndrome: plasma leakage → circulatory collapse.
DIAGNOSIS	RT-PCR, serology	
TREATMENT	Supportive. Steroids or disease-modifying antirheumatic drugs for chronic arthritis.	Supportive. Intravascular volume repletion or blood transfusion if severe shock.
PREVENTION	Minimize mosquito exposure. No vaccine currently available.	Live, recombinant vaccine available. Derived from the yellow fever virus backbone with insertion of genes for the envelope and pre-membrane proteins of dengue virus.

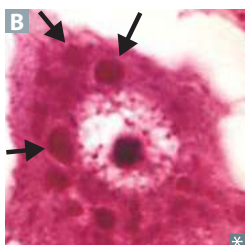
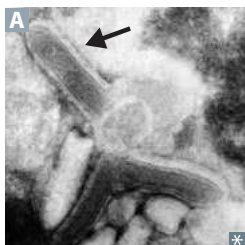
Yellow fever virus	A flavivirus (also an arbovirus) transmitted by <i>Aedes</i> mosquito bites. Virus has monkey or human reservoir. <i>Flavi</i> = yellow, jaundice. Symptoms: high fever, black vomitus, jaundice, hemorrhage, backache. May see Councilman bodies (eosinophilic apoptotic globules) on liver biopsy. Live, attenuated vaccine recommended for travelers to endemic countries.
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Zika virus



A flavivirus most commonly transmitted by *Aedes* mosquito bites.
Causes conjunctivitis, low-grade pyrexia, and itchy rash in 20% of cases. Outbreaks more common in tropical and subtropical climates. May be complicated by Guillain-Barré syndrome. Supportive care, no definitive treatment.
Diagnose with RT-PCR or serology.
Sexual and vertical transmission occurs.
In pregnancy, can lead to miscarriage or congenital Zika syndrome: brain imaging shows ventriculomegaly, subcortical calcifications **A**. Clinical features in the affected newborn include

- Microcephaly
- Ocular anomalies
- Motor abnormalities (spasticity, seizures)

Rabies virus

Bullet-shaped virus **A**. Negri bodies (cytoplasmic inclusions **B**) commonly found in Purkinje cells of cerebellum and in hippocampal neurons. Rabies has long incubation period (weeks to months) before symptom onset. Postexposure prophylaxis is wound cleaning plus immunization with killed vaccine and rabies immunoglobulin. Example of passive-active immunity.

Travels to the CNS by migrating in a retrograde fashion (via dynein motors) up nerve axons after binding to ACh receptors.

Progression of disease: fever, malaise
→ agitation, photophobia, hydrophobia, hypersalivation → paralysis, coma → death.

Infection more commonly from bat, raccoon, and skunk bites than from dog bites in the United States; aerosol transmission (eg, bat caves) also possible.

Ebola virus

A filovirus **A**. Following an incubation period of up to 21 days, presents with abrupt onset of flulike symptoms, diarrhea/vomiting, high fever, myalgia. Can progress to DIC, diffuse hemorrhage, shock.

Diagnosed with RT-PCR within 48 hr of symptom onset. High mortality rate.

Transmission requires direct contact with bodily fluids, fomites (including dead bodies), infected bats or primates (apes/monkeys); high incidence of healthcare-associated infection.

Supportive care, no definitive treatment.

Vaccination of contacts, strict isolation of infected individuals, and barrier practices for healthcare workers are key to preventing transmission.

Severe acute respiratory syndrome coronavirus 2

SARS-CoV-2 is a $\oplus$ ssRNA virus and the cause of the COVID-19 pandemic. Spread is via respiratory droplets, aerosols and fomites.

Varying symptoms include: fever, myalgia, headache, nasal congestion, sneezing, cough, sore throat, nausea, diarrhea, anosmia, dysgeusia.

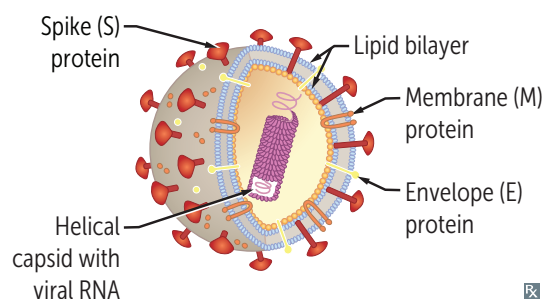
Severe cases may lead to complications like: pneumonia, hypercoagulability (DVT, PE, stroke), myocardial injury, neurologic sequelae, shock, organ failure, death.

Diagnosed by:

- NAAT (RT-PCR)
- Viral Antigen Test (RAT) → rapid, accessible, less sensitive than NAAT

Host cell entry occurs by attachment of viral spike protein to ACE2 receptor on cell membranes (highly expressed in the lungs, intestines, heart, and kidneys).

Vaccines: mRNA and viral vector vaccines induce immunity by presenting the spike protein to the immune system leading to anti-spike protein antibodies.

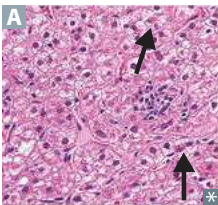


Hepatitis viruses

Signs and symptoms of all hepatitis viruses: episodes of fever, jaundice, ↑ ALT and AST. **NAkEd** viruses (HAV and HEV) lack an envelope and are not destroyed by the gut: the **vowels** hit your **bowels**.

HBV DNA polymerase has DNA- and RNA-dependent activities. Upon entry into nucleus, the polymerase completes the partial dsDNA. Host RNA polymerase transcribes mRNA from viral DNA to make viral proteins. The DNA polymerase then reverse transcribes viral RNA to DNA, which is the genome of the progeny virus.

HCV lacks 3'-5' exonuclease activity → no proofreading ability → antigenic variation of HCV envelope proteins. Host antibody production lags behind production of new mutant strains of HCV.

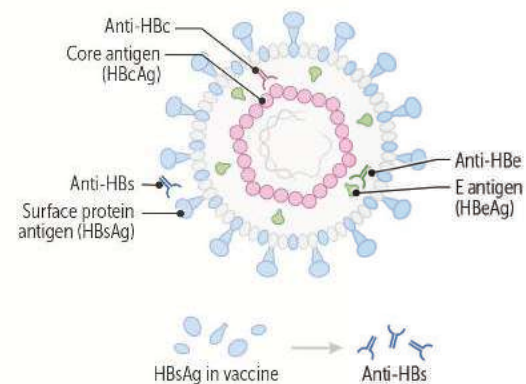
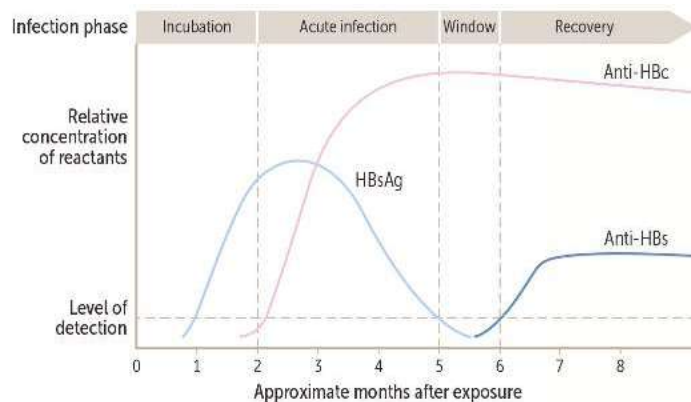
Virus	HAV	HBV	HCV	HDV	HEV
FAMILY	RNA picornavirus	DNA hepadnavirus	RNA flavivirus	RNA deltavirus	RNA hepevirus
TRANSMISSION	Fecal-oral (shellfish, travelers, day care)	Parenteral (B lood), sexual (B edroom), perinatal (B irthing)	Primarily blood (injection drug use, posttransfusion)	Parenteral, sexual, perinatal	Fecal-oral, especially waterborne
INCUBATION	Short (weeks)	Long (months)	Long	Superinfection (HDV after HBV) = short Coinfection (HDV with HBV) = long	Short
CLINICAL COURSE	A cute and self limiting (adults), A symptomatic (children)	Initially like serum sickness (fever, arthralgias, rash); may progress to carcinoma	May progress to C irrhosis or C arcinoma	Similar to HBV	Fulminant hepatitis in E xpectant (pregnant) patients
PROGNOSIS	Good	Adults → mostly full resolution; neonates → worse prognosis	Majority develop stable, C hronic hepatitis C	Superinfection → worse prognosis	High mortality in pregnant patients
HCC RISK	No	Yes	Yes	Yes	No
LIVER BIOPSY	Hepatocyte swelling, monocyte infiltration, Councilman bodies A 	Granular eosinophilic “ground glass” appearance due to accumulation of surface antigen within infected hepatocytes; cytotoxic T cells mediate damage	Lymphoid aggregates with focal areas of macrovesicular steatosis	Similar to HBV	Patchy necrosis
NOTES	A bsent (no) carrier state	Carrier state common	C arrier state very common	D efective virus, D epends on HBV HBsAg coat for entry into hepatocytes	E nteric, E pidemic (eg, in parts of Asia, Africa, Middle East), no carrier state

Extrahepatic manifestations of hepatitis B and C

	Hepatitis B	Hepatitis C
HEMATOLOGIC	Aplastic anemia	Essential mixed cryoglobulinemia, ↑ risk B-cell NHL, ITP, autoimmune hemolytic anemia
RENAL	Membranous GN > membranoproliferative GN	Membranoproliferative GN > membranous GN
VASCULAR	Polyarteritis nodosa	Leukocytoclastic vasculitis
DERMATOLOGIC		Sporadic porphyria cutanea tarda, lichen planus
ENDOCRINE		↑ risk of diabetes mellitus, autoimmune hypothyroidism

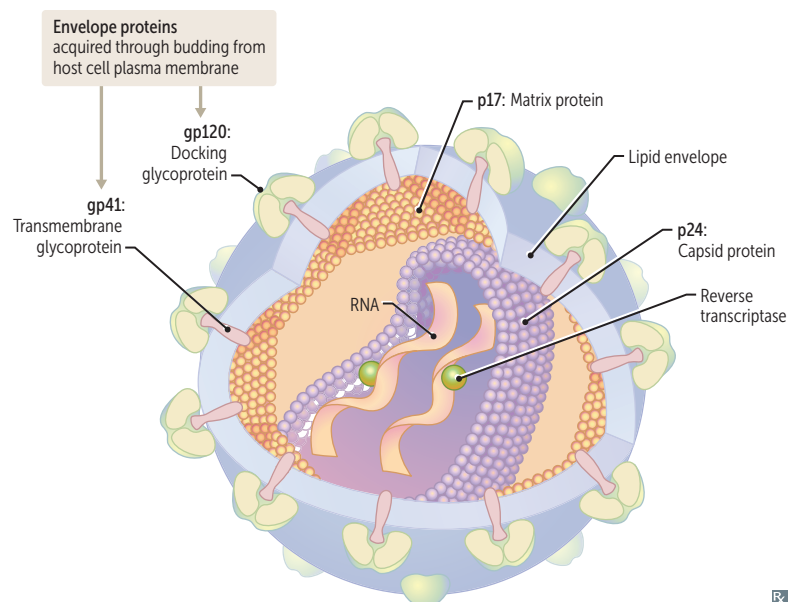
Hepatitis serologic markers

Anti-HAV (IgM)	IgM antibody to HAV; best test to detect acute hepatitis A.
Anti-HAV (IgG)	IgG antibody indicates prior HAV infection and/or prior vaccination; protects against reinfection.
HBsAg	Antigen found on surface of HBV; indicates hepatitis B infection.
Anti-HBs	Antibody to HBsAg; indicates immunity to hepatitis B due to vaccination or recovery from infection.
HBcAg	Antigen associated with core of HBV.
Anti-HBc	Antibody to HBcAg; IgM = acute/recent infection; IgG = prior exposure or chronic infection. IgM anti-HBc may be the sole ⊕ marker of infection during window period.
HBeAg	Secreted by infected hepatocyte into circulation. Not part of mature HBV virion. Indicates active viral replication and therefore high transmissibility and poorer prognosis.
Anti-HBe	Antibody to HBeAg; indicates low transmissibility.



	HBsAg	Anti-HBs	Anti-HBc	HBeAg	Anti-HBe
Incubation	+				
Acute infection	+		+ (IgM)	+	
Window			+ (IgM)		+
Recovery		+	+ (IgM)		+
Chronic infection (high infectivity)	+		+ (IgG)	+	
Chronic infection (low infectivity)	+		+ (IgG)		+
Immunized		+			

HIV



Diploid genome (2 molecules of RNA).

The 3 structural genes (protein coded for):

- *Env* (gp120 and gp41)—formed from cleavage of gp160 to form envelope glycoproteins.
 - gp120—attachment to host CD4+ T cell.
 - gp41 (forty-one)—fusion and entry.
- *gag* (p24 and p17)—capsid and matrix proteins, respectively.
- *pol*—Reverse transcriptase, Integrase, Protease; RIP “Pol” (Paul)

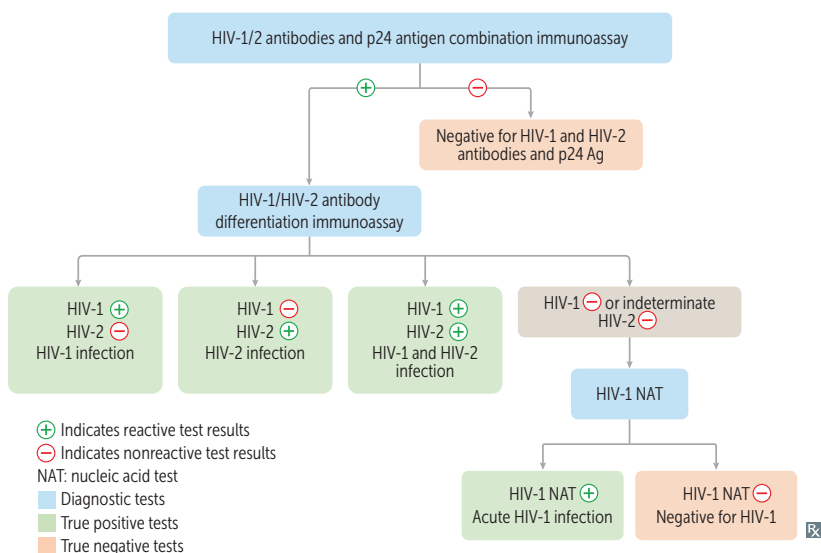
Reverse transcriptase synthesizes dsDNA from genomic RNA; dsDNA integrates into host genome.

Virus binds CD4 as well as a coreceptor, either CCR5 on macrophages (early infection) or CXCR4 on T cells (late infection).

Homozygous CCR5 mutation = immunity.

Heterozygous CCR5 mutation = slower course.

HIV diagnosis



HIV-1/2 Ag/Ab immunoassays detect viral p24 antigen capsid protein and IgG and/or IgM to HIV-1/2.

- Use for diagnosis. Very high sensitivity/specificity, but may miss early HIV disease if tested within first 2 weeks of infection.
- A positive screening test is followed by a confirmatory HIV-1/2 differentiation immunoassay.

HIV RNA tests detect elevated HIV RNA and can be qualitative or quantitative.

- NAAT is qualitative, and is a sensitive method to detect HIV viremia in antibody-negative patients.
- Viral load tests (RT-PCR) are quantitative and determine amount of viral RNA in the plasma. Use to monitor response to treatment and transmissibility.

Western blot tests are no longer recommended by the CDC for confirmatory testing.

HIV-1/2 Ag/Ab testing is not recommended in babies with suspected HIV due to maternally transferred antibody. Use HIV viral load instead.

AIDS diagnosis: ≤ 200 CD4+ cells/mm³ (normal: 500–1500 cells/mm³) or HIV ⊕ with AIDS-defining condition (eg, *Pneumocystis pneumonia*).

Common diseases of HIV-positive adults

↓ CD4+ cell count → reactivation of past infections (eg, TB, HSV, shingles), dissemination of bacterial infections and fungal infections (eg, coccidioidomycosis), and non-Hodgkin lymphomas.

PATHOGEN	PRESENTATION	FINDINGS
CD4+ cell count < 500/mm³		
<i>Candida albicans</i>	Oral thrush	Scrapable white plaque, pseudohyphae on microscopy
EBV	Oral hairy leukoplakia	Unscrapable white plaque on lateral tongue
HHV-8	Kaposi sarcoma, localized cutaneous disease	Perivascular spindle cells invading and forming vascular tumors on histology
HPV	Squamous cell carcinoma at site(s) of sexual contact (most commonly anus, cervix, oropharynx)	
<i>Mycobacterium tuberculosis</i>	Increased risk of reactivation of latent TB infection	
CD4+ cell count < 200/mm³		
<i>Histoplasma capsulatum</i>	Fever, weight loss, fatigue, cough, dyspnea, nausea, vomiting, diarrhea	Oval yeast cells within macrophages
HIV	Dementia, HIV-associated nephropathy	Cerebral atrophy on neuroimaging
JC virus (reactivation)	Progressive multifocal leukoencephalopathy	Nonenhancing areas of demyelination on MRI
HHV-8	Kaposi sarcoma, disseminated disease (pulmonary, GI, lymphatic)	
<i>Pneumocystis jirovecii</i>	<i>Pneumocystis</i> pneumonia	“Ground-glass” opacities on chest imaging
CD4+ cell count < 100/mm³		
<i>Bartonella</i> spp	Bacillary angiomatosis	Multiple red to purple papules or nodules Biopsy with neutrophilic inflammation
<i>Candida albicans</i>	Esophagitis	White plaques on endoscopy; yeast and pseudohyphae on biopsy
CMV	Colitis, Retinitis, Esophagitis, Encephalitis, Pneumonitis (CREEP)	Linear ulcers on endoscopy, cotton-wool spots on fundoscopy Biopsy reveals cells with intranuclear (owl eye) inclusion bodies
<i>Cryptococcus neoformans</i>	Meningitis	Encapsulated yeast on India ink stain or capsular antigen ⊕
<i>Cryptosporidium</i> spp	Chronic, watery diarrhea	Acid-fast oocysts in stool
EBV	B-cell lymphoma (eg, non-Hodgkin lymphoma, CNS lymphoma)	CNS lymphoma—ring enhancing, may be solitary (vs <i>Toxoplasma</i>)
<i>Mycobacterium avium-intracellulare</i> , <i>Mycobacterium avium</i> complex	Nonspecific systemic symptoms (fever, night sweats, weight loss, diarrhea) or superficial lymphadenitis	Most common if CD4+ cell count < 50/mm ³
<i>Toxoplasma gondii</i>	Brain abscesses	Multiple ring-enhancing lesions on MRI

Prions

Prion diseases are caused by the conversion of a normal (predominantly α -helical) protein termed prion protein (PrP^c) to a β -pleated form (PrP^{sc}), which is transmissible via CNS-related tissue (iatrogenic CJD) or food contaminated by BSE-infected animal products (variant CJD). PrP^{sc} resists protease degradation and facilitates the conversion of still more PrP^c to PrP^{sc}. Resistant to standard sterilizing procedures, including standard autoclaving. Accumulation of PrP^{sc} results in spongiform encephalopathy and dementia, ataxia, startle myoclonus, and death.

Creutzfeldt-Jakob disease—rapidly progressive dementia, typically sporadic (some familial forms).

Bovine spongiform encephalopathy—also called “mad cow disease.”

Kuru—acquired prion disease noted in tribal populations practicing human cannibalism.

▶ MICROBIOLOGY—SYSTEMS

Normal microbiota:
dominant

Neonates delivered by C-section have microbiota enriched in skin commensals.

LOCATION	MICROORGANISM
Skin	<i>S epidermidis</i>
Nose	<i>S epidermidis</i> ; colonized by <i>S aureus</i>
Oropharynx	Viridans group streptococci
Dental plaque	<i>S mutans</i>
Colon	<i>B fragilis</i> > <i>E coli</i>
Vagina	<i>Lactobacillus</i> ; colonized by <i>E coli</i> and group B strep

Bugs causing food-
borne illness

S aureus and *B cereus* food poisoning starts quickly and ends quickly (exotoxin-mediated).

MICROORGANISM	SOURCE OF INFECTION
<i>B cereus</i>	Reheated rice. “Food poisoning from reheated rice? Be serious! ” (<i>B cereus</i>)
<i>C botulinum</i>	Improperly canned foods (toxins), raw honey (spores)
<i>C perfringens</i>	Reheated meat
<i>E coli</i> O157:H7	Undercooked meat
<i>L monocytogenes</i>	Deli meats, soft cheeses
<i>Salmonella</i>	Poultry, meat, and eggs
<i>S aureus</i>	Meats, mayonnaise, custard; preformed toxin
<i>V parahaemolyticus</i> and <i>V vulnificus</i> ^a	Raw/undercooked seafood

^a*V vulnificus* predominantly causes wound infections from contact with contaminated water or shellfish.